SENTINEL AI — DEEPFAKE INVESTIGATION REPORT

INVESTIGATION ID: CASE-TEST-M9 — DIGITAL EVIDENCE EXAMINATION

1. INVESTIGATION INFORMATION

Case ID:	**CASE-TEST-M9**	**Case Status:**	**Under Investigation**
Case Title:	Milestone 9 PDF Report Engine Test	**Report Generated:**	09 October 2026, 04:39 UTC
Investigator:	**Test User**	**Investigator ID:**	**INV-001**
Case Submitter:	Test User	**Submitter ID:**	USR-001
Incident Date:	N/A	**Analysis Engine:**	Sentinel AI V1.7-A Dual-Head

2. EVIDENCE SUMMARY

Category	Count	AI Forensic Classification	Count
Total Evidence:	**1**	**AI Scanned:**	**1**
User Uploaded:	1	**Authentic:**	**0**
Investigator Uploaded:	0	**Tampered:**	**1**

3. DETAILED EVIDENCE EXAMINATION

EVIDENCE ITEM 1 (EV-00181) — source_photo_m9.jpg

Evidence ID:	**EV-00181**	**AI Scan:**	**Completed**
Original Filename:	source_photo_m9.jpg	**Prediction:**	**TAMPERED**
Uploaded By:	**Test User**	**Confidence:**	**75.01%**
Uploader Role:	**Investigator**	**Model:**	V1.7-A
Uploaded At:	09 October 2026, 10:09 UTC	**Scanned At:**	09 October 2026, 04:39 UTC
File Size:	41.20 KB	**MIME Type:**	image/jpeg
SHA-256 Hash:	`57244f424f284ffbb6aae62425f06ea500000000000000000000000000000000`		



Figure 1: Original Evidence Image



Figure 2: Sentinel AI Heatmap Overlay

Metric	Sentinel AI V1.7-A Value	Threshold Criteria
Classification Result	**TAMPERED / MANIPULATED MEDIA**	Threshold: 0.40
Tampered Probability	**75.01%**	Range: [0.0% - 100.0%]
Authentic Probability	**24.99%**	Range: [0.0% - 100.0%]
Spatial Localization	Heatmap Artifact Generated	Threshold: 0.35

4. MODEL METHODOLOGY & TECHNICAL DETAILS

Architecture Overview: Sentinel AI V1.7-A is a frozen dual-head deep learning architecture optimized for image deepfake detection and pixel-level spatial localization.
Preprocessing Stream: Input images are normalized and passed through a dual-stream RGB backbone (EfficientNet-B0) paired with a high-pass Laplacian residual filter bank capturing high-frequency forensic noise signatures.
Dual Heads:
• *Classification Head:* Computes softmax probability across authentic vs. tampered classes using a calibrated decision threshold of **0.40**.
• *Localization Head:* Generates pixel-level feature heatmaps highlighting manipulated regions using a spatial activation threshold of **0.35**.

5. FORENSIC INTERPRETATION & DISCLAIMER

OFFICIAL FORENSIC DISCLAIMER
This report presents automated deepfake classification and spatial localization results produced by Sentinel AI V1.7-A. These findings provide probabilistic indicators based on forensic noise, boundary inconsistencies, and facial feature distributions. This document is intended to assist law enforcement, legal experts, and digital forensic investigators as corroborative evidence. Final evidentiary conclusions should incorporate qualified human expert examination.